FOOD DELIVERY SERVICE DATA CLEANING REPORT

Comparison of Messy Data and Cleaned Data

Project report based on the two supplied Excel files: Messy_Data(2).xlsx and Cleaned_Data(2).xlsx.

1. Introduction

This report explains the data-cleaning process applied to a sample food-delivery dataset. The original worksheet contained inconsistent text formats, missing values, an invalid date, a negative quantity, and duplicate-looking records. The cleaned worksheet standardizes the data and adds a calculated order total to support basic analysis.

2. Objectives

- Identify data-quality problems in the original spreadsheet.
- Standardize city names, order statuses, and date formats.
- Handle missing and invalid values and review duplicate records.
- Calculate each order's total amount and summarize the cleaned data.

3. Dataset Overview

Measure	Messy dataset	Cleaned dataset
Number of records	15	13
Fields/columns	9	10
Date range represented	1–8 October 2026 (plus one invalid date)	1–8 October 2026
Additional calculated field	Not available	Total_Amount_INR

4. Data-Quality Issues Identified

Issue	Example in original data	Cleaning action reflected in cleaned file
Inconsistent city spelling/case	chennai, CHENNAI, Madras	Standardized to Chennai
Inconsistent status case	delivered, DELIVERED, in transit	Standardized to Delivered / In Transit
Mixed date formats	2026-10-01, 01/10/2026, 2026/10/02	Converted to YYYY-MM-DD
Missing values	Quantity missing for order 1005; unit price missing for order 1006	Filled with 2 and 60 respectively
Invalid date	Order 1011 has 'not-a-date'	Set to 2026-10-06 in cleaned file
Negative quantity	Order 1012 has quantity -1	Corrected to quantity 2
Duplicate-looking rows	Order 1010 repeats order 1001; order 1013 resembles order 1009	Records reduced from 15 to 13

Important validation note: The cleaned file assumes the missing quantity, missing unit price, invalid date, and negative quantity can be corrected using the values shown. These assumptions should be checked against original order records before the data is used for business decisions. Order 1013 has a different Order_ID from 1009, so it should only be removed if the source system confirms it is a duplicate.

5. Cleaning and Transformation Steps

- Normalized city names so Chennai variants and Madras use the same city label.
- Standardized order status capitalization and wording.
- Converted dates into a consistent YYYY-MM-DD format and replaced the invalid date using the value in the cleaned file.
- Filled missing Quantity and Unit_Price_INR values and corrected the negative quantity.
- Removed two duplicate-looking records to produce 13 rows.

• Added Total_Amount_INR using Quantity × Unit_Price_INR.

6. Summary of the Cleaned Dataset

Metric	Result
Cleaned orders	13
Total order value (all statuses)	■5,440
Average order value	■418.46
Delivered orders	11
Cancelled orders	1
In-transit orders	1

7. Product and City Analysis

Food item	Orders	Total quantity	Order value
Pizza	4	5	■2,250
Biryani	3	7	■1,750
Burger	3	5	■900
Sandwich	1	2	■300
Dosa	2	4	■240

Pizza has the highest order value at ■2,250, followed by Biryani at ■1,750. Chennai has the largest number of orders (6) and the highest city-level order value (■2,790). These totals include cancelled and in-transit orders because the report sums all rows in the cleaned sheet; they should not be interpreted as completed revenue.

8. Tools and Formula

Tool used: Microsoft Excel. The cleaned worksheet adds a calculated field with the formula: **Total_Amount_INR = Quantity × Unit_Price_INR**. Example: 2 pizzas × ■450 = ■900.

9. Conclusion

The cleaning process improves consistency and makes the food-delivery data easier to analyze. The cleaned dataset contains 13 records, standardized city/status/date values, corrected quantities and prices, and a calculated total amount for each order. Before final use, the assumptions used to repair invalid or missing values and remove duplicate-looking records should be verified against the original order system.